



RIVER VALLEY HIGH SCHOOL

YEAR 6 PRELIMINARY EXAMINATION

CANDIDATE
NAME

CLASS

6	
---	--

CENTRE
NUMBER

S				
---	--	--	--	--

INDEX
NUMBER

--	--	--	--

H1 CHEMISTRY

8872/02

Paper 2

13 September 2011

2 hours

Candidates answer Section A on the Question Paper.

Additional Materials: Data Booklet

READ THESE INSTRUCTIONS FIRST

Write your name, class, Centre number and index number on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** the questions.

Section B

Answer **two** questions on separate answer paper.

A Data Booklet is provided. Do not write anything on it.

The number of marks is given in brackets [] at the end of each question or part question.
At the end of the examination, fasten all your work securely together.

For Examiner's Use	
Section A	
B5	
B6	
B7	
Total	

This document consists of **13** printed pages.

BLANK PAGE

Formatted: Font: 24 pt, Complex Script Font:
24 pt

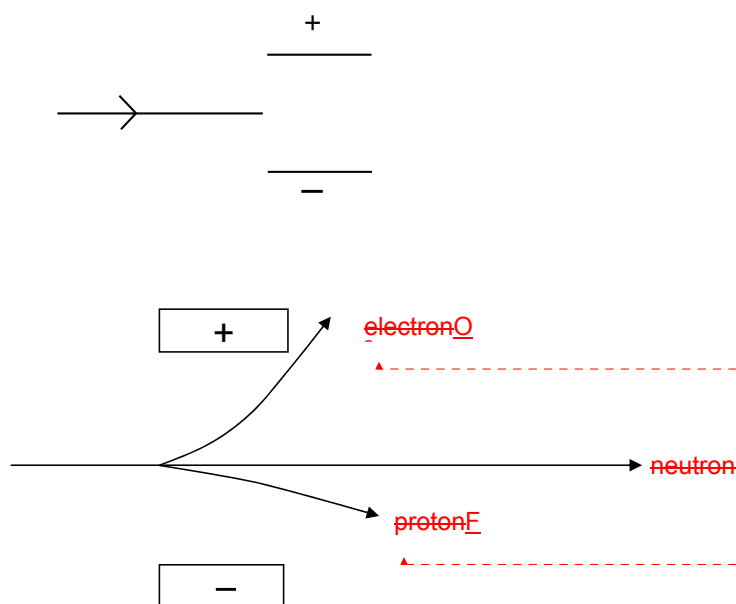
Section A

Answer **all** questions in this section.

1. (a) Complete the table below.

Element	Electric charge	Mass number	Number of		
			Protons	Electrons	Neutrons
N	0	14	7	7	7
O	2-	18	8	10	10
F	1+	19	9	8	10

- (b) Beams consisting of the particles listed in the table above are subjected to an electric field as shown in the diagram below. Sketch, and label, on the diagram below to show how beams of each of the three particles are affected by the electric field.



- (c) (i) Write an equation to define the first ionisation energy of nitrogen.



- (ii) Sketch a graph of the first ionisation energy of the elements of Period 2.

Formatted: Font: 24 pt, Complex Script Font: 24 pt

Formatted: Highlight

Formatted: Highlight

Formatted: Highlight

Formatted: Highlight

Formatted: Highlight

Formatted Table

Formatted: Justified, Space Before: 6 pt, After: 6 pt

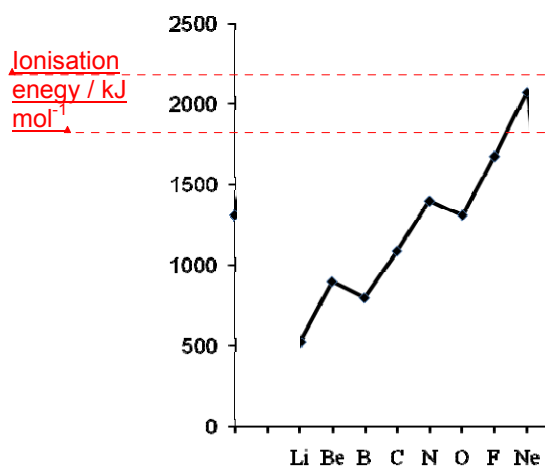
Formatted: Superscript

Formatted: Superscript

Formatted: Highlight

Formatted: Highlight

Formatted: Highlight



Formatted: Font: (Default) Arial, Complex
Script Font: Arial

Formatted: Font: (Default) Arial, Complex
Script Font: Arial, Superscript

(iii) Explain the difference between the first ionisation energy of

I nitrogen and oxygen

II oxygen and fluorine

[5]

Electrostatic repulsion between the paired 2p electrons in O makes it easier to remove one of the paired 2p electrons, than the unpaired 2p electrons from N.

Hence, the 1st IE of O is smaller than the 1st IE of N.

Formatted: No underline, Highlight

Formatted: Indent: Left: 0", Tab stops: Not at 3" + 6"

Formatted: Highlight

F has a large first ionisation energy than O as F has a larger nuclear charge than O while there is negligible increase in screening effect as electrons are added to the same shell. The atomic radius of F is also smaller than O.

[Total:10]

- 2 Chlorine is a pale green gas at room temperature and pressure. There are two stable isotopes of chlorine, ^{35}Cl and ^{37}Cl . In addition, a radioactive isotope, ^{36}Cl , exists in trace amounts in the environment and can be used for radioactive dating.

Chlorine is toxic. It can react with water to form a mixture of hydrochloric acid and hypochlorous acid, HOCl . As such, it causes irritation to the respiratory system if inhaled as it can dissolve in moist surfaces of the air passages and the lungs.

Chlorine can be detected by smell at 3 ppm or parts per million ($1 \text{ ppm} = 1 \text{ mg dm}^{-3}$). At a concentration of 60 ppm, lung damage may occur. Death can result from inhaling chlorine gas at levels above 1000 ppm.

- (a) A sample of chlorine gas consisting of the isotopes ^{35}Cl and ^{37}Cl was found to have a relative atomic mass of 35.5. This sample was subsequently contaminated with the radioactive isotope ^{36}Cl . The relative atomic mass of the chlorine in the contaminated sample was found to be 35.6. Find the percentage abundance of ^{36}Cl present in the contaminated sample.

Let x be the fraction of ^{35}Cl in the original (uncontaminated) sample.

$$35(x) + 37(1-x) = 35.5$$

$$x = 0.75$$

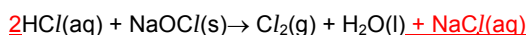
Let y be fraction of ^{36}Cl in the contaminated sample.

$$36(y) + 0.75(35) + 0.25(37) = 35.6$$

$$y = 0.002777$$

% abundance of ^{36}Cl present in the contaminated sample = 0.278%

- (b) Chlorine is a strong oxidising agent and is often used as a disinfectant in swimming pools. Due to the danger and inconvenience of storing pure chlorine, chlorine that is used for disinfecting swimming is often produced by reacting controlled amounts of hydrochloric acid with sodium hypochlorite.



In an accident, 2 dm^3 of 10 mol dm^{-3} $\text{HCl}(\text{aq})$ was mixed with 1 kg of solid NaOCl . The incident occurred in a room of volume 50 m^3 . The room was evacuated and equipment was brought in to ventilate the room. The equipment was able to ventilate the building at a rate of $5 \text{ m}^3 \text{ min}^{-1}$.

Formatted: Font color: Auto, Highlight

Formatted: Highlight

Formatted: Font color: Auto, Highlight

Formatted: Highlight

Formatted: No underline, Highlight

Formatted: Highlight

Formatted: Space Before: 0 pt, After: 0 pt, Line spacing: Multiple 1.15 li

Formatted Table

[3]

Formatted: Highlight

Formatted: Font: (Default) Arial, 11 pt, Complex Script Font: Arial, 11 pt, Highlight

Formatted: Highlight

Formatted: Font: (Default) Arial, 11 pt, Complex Script Font: Arial, 11 pt, Highlight

Formatted: Highlight

Formatted: Portuguese (Brazil)

Formatted: English (U.S.)

(i) Determine the oxidation number of chlorine in NaOCl.

+1

(ii) Calculate the amount of chlorine gas produced.

Amount of HCl(aq) = $2 \times 10 = 20 \text{ mol}$

Amount of NaOCl = $1000 / (23.0 + 16.0 + 35.5) = 13.4 \text{ mol}$

$2\text{HCl} \rightleftharpoons \text{NaOCl}$

Limiting reagent is HCl

Amount of $\text{Cl}_2 = 20 / 2 = 10 \text{ mol}$

(iii) Determine if the concentration of chlorine in the room is greater than the lethal concentration of 1000 ppm.

Concentration of chlorine

$= 10 / (50 \times 1000)$

$= 0.000200 \text{ mol dm}^{-3}$

$= 0.0142 \text{ g dm}^{-3}$

$= 14.2 \text{ mg dm}^{-3}$

Concentration of chlorine in the room is less than the lethal concentration.

(iv) Determine the theoretical volume of air needed to dilute the chlorine gas to a concentration lower than 3 ppm.

Mass of chlorine present = $10 \times 71 = 710 \text{ g} = 710 \times 10^3 \text{ mg}$

Volume of air required to dilute concentration to lower than 3 ppm

$= 710 \times 10^3 / 3$

$= 2.37 \times 10^6 \text{ dm}^3$

$= 2370 \text{ m}^3$

(v) The company supplying the ventilation equipment claimed that the equipment would be able to remove the chlorine gas from the room after ten minutes. Suggest if this is a realistic claim.

No, volume of air required to dilute the chlorine is too large to be pumped in 10 minutes.

[Total: 10]

Formatted: Font: Bold

Formatted: English (U.S.)

Formatted: Highlight

Formatted: English (U.S.)

Formatted: Font: Bold

Formatted: English (U.S.)

Formatted: Highlight

Formatted: Highlight

Formatted: Highlight

Formatted: Subscript, Highlight

Formatted: Highlight

Formatted: English (U.S.)

Formatted: Font: Bold

Formatted: English (U.S.)

Formatted: Highlight

Formatted: English (U.S.)

Formatted: Font: Bold

Formatted: Highlight

Formatted: Font: Bold

Formatted: Highlight

Formatted Table

Formatted: Font: Bold

- 3 Many biological processes only occur within a narrow range of pH values. The pH of different fluids found in the body is given below:

Saliva	pH 6.8
Blood	pH 7.4
Stomach juices	pH 1.0 to pH 3.0
Intestinal juices	pH 8.5

- (a) Calculate the hydroxide ion concentration in intestinal juices.

$$[H^+] = 10^{-8.5} = 3.16 \times 10^{-9}$$

$$[OH^-] = 10^{-14} / 3.16 \times 10^{-9} = 3.16 \times 10^{-6} \text{ mol dm}^{-3}$$

Formatted: Portuguese (Brazil)

Formatted: Portuguese (Brazil), Superscript

Formatted: Portuguese (Brazil)

Formatted: Portuguese (Brazil)

[2]

- (b) The pH of stomach juices is low due to the presence of hydrochloric acid. Hydrochloric acid can be classified as a *strong Bronsted acid*. Explain what you understand by the terms in italics.

[2]

A strong Bronsted acid is a substance that disassociates fully to produce

H^+

...

Formatted: Highlight

- (c) The body maintains the pH of blood within a narrow range of values. Death could result if the blood pH decreases below 6.8 or rises above 7.8. The need to maintain the pH within a narrow range of values requires the use of a buffer. In blood, the main buffering system is the H_2CO_3 / HCO_3^- buffer.

- (i) Explain what is meant by the term *buffer*.

A buffer solution is one that can resist a change in pH (pH changes only very slightly) when a small amount of acid or base is added to it.

Formatted: Font: Bold

Formatted: Font: Bold

Formatted: Highlight

Formatted: Font: Bold

Formatted: Font: Bold

Formatted: Font: Bold

Formatted: Highlight

Formatted: Left

Formatted: Highlight

Formatted: Subscript

Formatted: Highlight

Formatted: Highlight

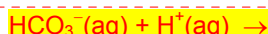
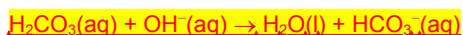
Formatted: Highlight

Formatted: Highlight

Formatted: Font: Bold

Formatted: Font: Bold

- (ii) Write equations to show how the H_2CO_3 / HCO_3^- buffer functions.



- (iii) During heavy exercise, acidic waste products are produced in the body. One function of blood is to remove such waste products. One such

[6]

response is an increased rate of breathing. In addition to increasing the supply of oxygen to the body, heavy breathing helps to remove a gas which in turn helps to regulate the pH of the blood.

State the identity of the gas removed and suggest an explanation how heavy breathing helps to regulate blood pH. Include an equation in your answer.

The gas is carbon dioxide.

Formatted: Highlight

Heavy breathing removes carbon dioxide from the body which shifts equilibrium (*) to the left, thus removing carbonic acid from the body in turn.

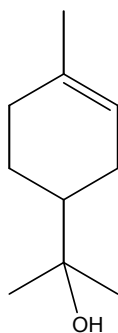


Formatted: Portuguese (Brazil)

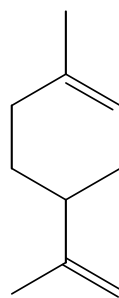
[Total: 10]

Formatted: Font: Bold

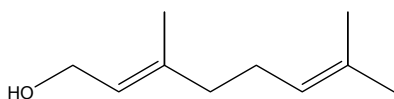
- 4 Terpeneol, limonene, citral and geraniol are members of a class of compounds called terpenes which are widely found in nature.



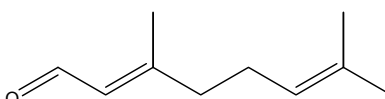
terpineol



limonene



geraniol



citral

- (a) Identify the functional groups present in geraniol and citral. [2]

geraniol :

primary alcohol, alkene

Formatted: Highlight

citral:

aldehyde, alkene

Formatted: Highlight

- (b) State the reagents and conditions by which you could convert geraniol to citral. [1]

acidified potassium dichromate, distil

Formatted: Highlight

- (c) Describe, by means of simple chemical tests, how you would distinguish between the following pairs of compounds.

- (i) geraniol and citral

Add Tollens reagent, heat.

Silver mirror formed for citral but not for geraniol.

OR

Add PCl_5 .

White fumes of HCl observed for geraniol but not for citral.

Formatted: Highlight

- (ii) terpineol and limonene [4]

Add PCl_5 .

White fumes of HCl observed for terpineol but not for limonene.

Formatted: Highlight

OR

Add acidified potassium manganate(VII), heat, and pass any gaseous products through limewater.

Colourless gas that gives a white precipitate with limewater produced for limonene but not for terpineol.

Formatted: Highlight

- (c) Both terpineol and limonene are liquids at room temperature. State and explain if terpineol has a higher boiling point than limonene.

[3]

Terpineol has a higher boiling point than limonene. More energy is needed to overcome the stronger hydrogen bonding between terpineol molecules.....than the weaker van der Waals forces between limonene molecules.....

Formatted: Highlight

Formatted: Justified

[Total: 10]

Formatted: Font: Bold

Formatted: Right: 0.15"

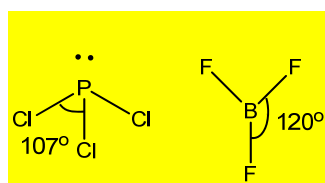
Section B

Answer **two** of the three questions in this section on separate answer paper.

5. Phosphorus forms three chlorides, P_2Cl_4 , PCl_3 and PCl_5 , of which the latter two are most common. PCl_3 is made industrially on a large scale by the direct chlorination of phosphorus, while PCl_5 serves as a valuable chemical intermediate in many processes.

[8]

- (a) (i) PCl_3 can function as a Lewis base, reacting with boron trifluoride, BF_3 , in the molar ratio of 1:1. Using the Valence Shell Electron Pair Theory, draw diagrams to illustrate the molecular shapes of PCl_3 and BF_3 and indicate the bond angles.

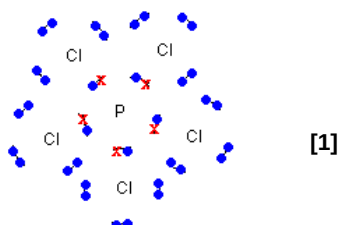


[1] for each correct structure

- (ii) Suggest the type of bonding that can take place between PCl_3 and BF_3 .

Dative/coordinate bonding. [1]

- (iii) Draw a dot-and-cross diagram of phosphorus pentachloride, PCl_5 , predicting the shape of the molecule.



[1]

Shape: Trigonal bipyramidal [1]

- (iv) PCl_5 is molecular in the gaseous phase. However, it adopts an ionic structure in the crystalline phase, with the lattice consisting of $[PCl_4]^+$ and $[PCl_6]^-$ ions.

Suggest a likely identity of the ion $[PCl_6]^-$.

A likely identity of the ion is $[PCl_6]^-$. [1]

- (v) When dissolved in water, PCl_5 gives a strongly acidic solution. Explain why this happens and provide a relevant equation.

PCl_5 undergoes hydrolysis in water to give strongly acidic solutions (pH 2). P uses its energetically accessible vacant 3d-orbitals for dative bonding with water molecules. [1]



Elements from Period 3 form oxides that exhibit different physical and chemical properties.

Compound	Na ₂ O	MgO	SiO ₂	P ₄ O ₆
Melting point / °C	1132	2800	1700	24

[4]

- (b) (i) Explain why MgO has a higher melting point than Na₂O in terms of structure and bonding.

Both MgO and Na₂O have giant ionic structures[✓] and strong electrostatic forces of attraction exist between their ions.[✓] Due to the higher charge density of Mg²⁺,[✓] more energy is needed[✓] to break the stronger electrostatic attraction between Mg²⁺ and O²⁻ ions, resulting in a higher melting point than Na₂O.

4[✓]: 2 mark

2-3[✓]: 1 mark

- (ii) Explain why P₄O₆ has a lower melting point than SiO₂ in terms of structure and bonding.

P₄O₆ has a simple covalent structure[✓] and weak van der Waals forces exist between its molecules.[✓] On the other hand, SiO₂ has a giant covalent structure[✓] and strong covalent bonds exist between its atoms.[✓] Thus, less energy[✓] is needed to break the van der Waals forces between P₄O₆ molecules, resulting in a lower melting point.

5[✓]: 2 marks

3-4[✓]: 1 mark

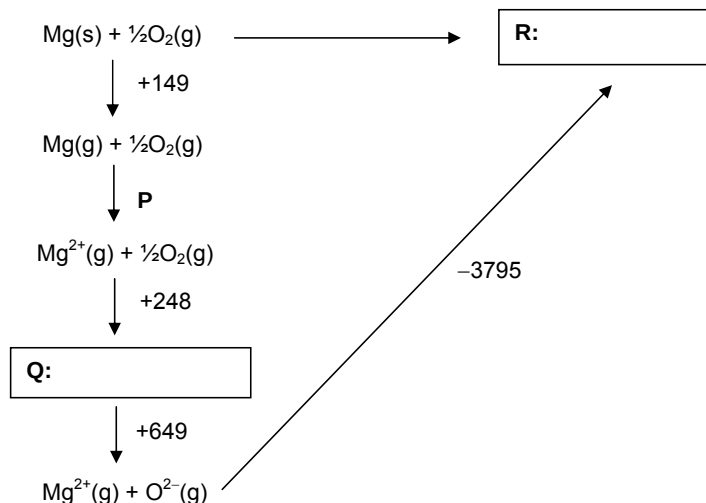
- (c) (i) Define standard enthalpy change of formation.

[5]

The standard enthalpy change of formation of a substance is the enthalpy change when one mole of the substance is formed from its constituent elements in their standard states (under standard conditions). [1]

- (ii) Using relevant data from the *Data Booklet*, complete the energy cycle by stating **P** to **R**. Include state symbols or units in your answers when necessary.

Therefore, calculate the enthalpy change of formation of magnesium oxide.



$$\text{P: } +2186 \text{ kJ mol}^{-1} \quad [1]$$

$$\text{Q: } \text{Mg}^{2+}\text{(g)} + \text{O(g)} \quad [1]$$

$$\text{R: } \text{MgO(s)} \quad [1]$$

$$\begin{aligned}
 \text{Enthalpy change of formation of MgO} &= +149 + 2186 + 248 + 649 - 3795 \\
 &= -563 \text{ kJ mol}^{-1} \quad [1]
 \end{aligned}$$

Liquid butane is commonly found in canisters used for portable gas stoves during camping trips.

- (d) When 1 g of butane was burnt under a container carrying 150 g of water, it was found that the temperature of water rose from 25 °C to an unknown temperature, T. Given that the enthalpy change of combustion of butane is -2877 kJ mol⁻¹ and the burning process is 80 % efficient, find the unknown temperature T. (Specific heat capacity of water = 4.2 J g⁻¹ K⁻¹) [3]

$$\begin{aligned}
 \text{Amount of limiting reagent (butane)} &= 1 / 58 \\
 &= 0.01724 \text{ mol} \quad [1]
 \end{aligned}$$

$$\begin{aligned}
 \text{When burning process is 100 \% efficient,} \\
 \text{Heat evolved} &= 150 \times 4.2 \times (T - 25) \times 10/8 \\
 &= 787.5 (T - 25) \text{ J} \quad [1]
 \end{aligned}$$

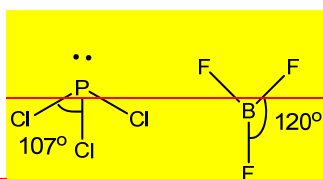
$$\begin{aligned}
 \Delta H_c(\text{butane}) &= \text{Heat evolved} / \text{amount of butane} \\
 -2877 \times 10^3 &= -787.5 (T - 25) / 0.01724 \\
 T &= 88.0 \text{ °C} \quad [1]
 \end{aligned}$$

[Total: 20]

1. Phosphorus forms three chlorides, P_2Cl_4 , PCl_3 and PCl_5 , of which the latter two are most common. PCl_3 is made industrially on a large scale by the direct chlorination of phosphorus, while PCl_5 serves as a valuable chemical intermediate in many processes.

[8]

- (a) (i) PCl_3 can function as a Lewis base, reacting with boron trifluoride, BF_3 , in the molar ratio of 1:1. Using the Valence Shell Electron Pair Theory, draw diagrams to illustrate the molecular shapes of PCl_3 and BF_3 and indicate the bond angles.

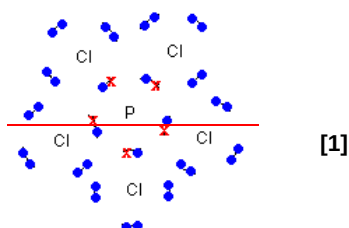


[1] for each correct structure

- (ii) Suggest the type of bonding that can take place between PCl_3 and BF_3 .

Dative/coordinate bonding. [1]

- (iii) Draw a dot and cross diagram of phosphorus pentachloride, PCl_5 , predicting the shape of the molecule.



[1]

Shape: Trigonal bipyramidal [1]

- (iv) PCl_5 is molecular in the gaseous phase. However, it adopts an ionic structure in the crystalline phase, with the lattice consisting of $[PCl_4]^+$ and $[PCl_6]^-$ ions.

Suggest a likely identity of the ion $[PCl_6]^-$.A likely identity of the ion is $[PCl_6]^-$. [1]

- (v) When dissolved in water, PCl_5 gives a strongly acidic solution. Explain why this happens and provide a relevant equation.

PCl_5 undergoes hydrolysis in water to give strongly acidic solutions (pH 2). P uses its energetically accessible vacant 3d-orbitals for dative bonding with water molecules. [1]



Elements from Period 3 form oxides that exhibit different physical and chemical properties.

Compound	Na ₂ O	MgO	SiO ₂	P ₄ O ₆
Melting point / °C	1132	2800	1700	24

[4]

- (b) (i) Explain why MgO has a higher melting point than Na₂O in terms of structure and bonding.

Both MgO and Na₂O have giant ionic structures and strong electrostatic forces of attraction exist between their ions. Due to the higher charge density of Mg²⁺, more energy is needed to break the stronger electrostatic attraction between Mg²⁺ and O²⁻ ions, resulting in a higher melting point than Na₂O.

4[✓]: 2 mark

2-3[✓]: 1 mark

- (ii) Explain why P₄O₆ has a lower melting point than SiO₂ in terms of structure and bonding.

P₄O₆ has a simple covalent structure and weak van der Waals forces exist between its molecules. On the other hand, SiO₂ has a giant covalent structure and strong covalent bonds exist between its atoms. Thus, less energy is needed to break the van der Waals forces between P₄O₆ molecules, resulting in a lower melting point.

5[✓]: 2 marks

3-4[✓]: 1 mark

Formatted Table

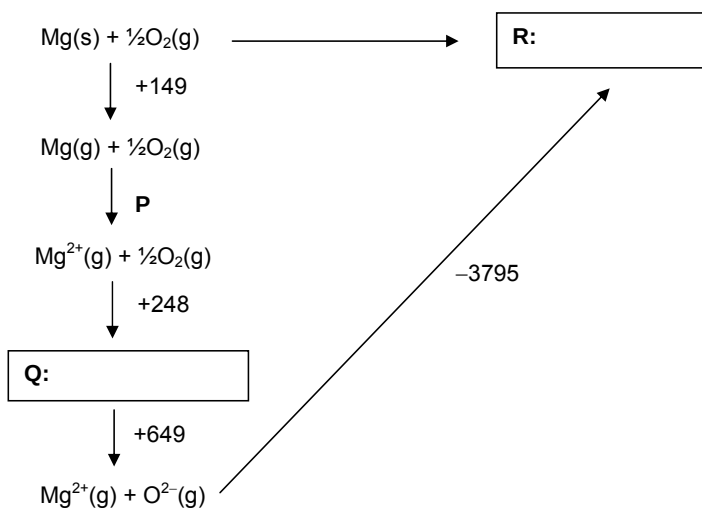
- (c) (i) Define standard enthalpy change of formation.

[8]

The standard enthalpy change of formation of a substance is the **enthalpy change** when **one mole** of the substance is **formed from its constituent elements in their standard states** (under standard conditions). [1]

- (ii) Using relevant data from the Data Booklet, complete the energy cycle by stating **P** to **R**. Include state symbols or units in your answers when necessary.

Therefore, calculate the enthalpy change of formation of magnesium oxide.



Formatted Table

$$\text{Enthalpy change of formation of MgO} = +149 + 2186 + 248 + 649 - 3795$$

$$= -563 \text{ kJ mol}^{-1} \quad [1]$$

Liquid butane is commonly found in canisters used for portable gas stoves during camping trips.

Formatted: Justified

- (d) When 1 g of butane was burnt under a container carrying 150 g of water, it was found that the temperature of water rose from 25 °C to an unknown temperature, **T**. Given that the enthalpy change of combustion of butane is $-2877 \text{ kJ mol}^{-1}$ and the burning process is 80 % efficient, find the unknown temperature **T**. (Specific heat capacity of water = $4.2 \text{ J g}^{-1} \text{ K}^{-1}$)

[3]

$$\text{Amount of limiting reagent (butane)} = 1 / 58$$

$$= 0.01724 \text{ mol} \quad [1]$$

$$\text{When burning process is 100 \% efficient,}$$

$$\text{Heat evolved} = 150 \times 4.2 \times (T - 25) \times 100/80$$

$$= 787.5 (T - 25) \text{ J} \quad [1]$$

$$\Delta H_c(\text{butane}) = \text{Heat evolved} / \text{amount of butane}$$

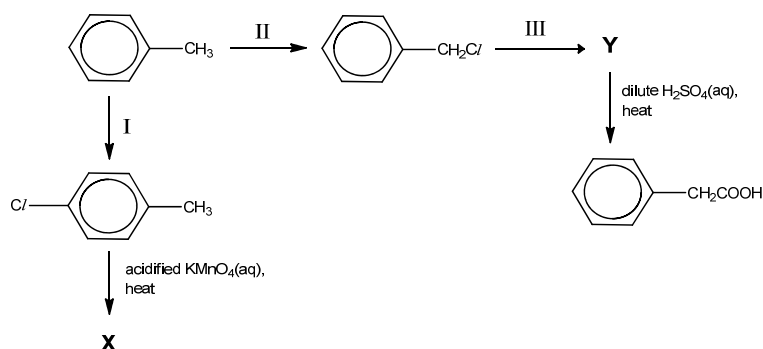
$$-2877 \times 10^3 = 787.5 (T - 25) / 0.01724$$

T = 88.0 °C [1]

[Total: 20]

Formatted: Font: Bold

6. (a) A reaction scheme starting from methylbenzene is shown below. Draw the structures of the organic products X and Y and state the reagents and conditions needed for steps I to III. [5]

Step I: Cl_2 , $\text{Fe}/\text{FeCl}_3/\text{AlCl}_3$ catalyst [1]Step II: Cl_2 , uv light [1]

Step III: ethanolic KCN, heat [1]

A:		[1]
B:		[1]

- (b) Rank the following compounds in order of increasing acidity.

Propanoic acid, propanol, 2-chloropropanol, 2-bromopropanoic acid

[4]

Acidity: Propanol < 2-chloropropanol < propanoic acid < 2-bromopropanoic acid [1]

Acid strength depends on stability of anion. The more stable the anion, the stronger the acid.

In 2-chloropropanol, the electron-withdrawing Cl atom disperses the negative charge on the alkoxide anion, [✓] stabilising the anion. On the other hand, in propanol, the electron-donating alkyl group intensifies the negative charge on the alkoxide anion, [✓] destabilising the anion.

In propanoic acid, the negative charge on the carboxylate anion is delocalised into the COO^- group, [✓] stabilising the anion to a larger

extent as compared to the alkoxide anion.[✓] In 2-bromopropanoic acid, the negative charge on the carboxylate anion is both delocalised into the COO^- group and dispersed by the electron-withdrawing Br atom,[✓] thus stabilising the anion to the largest extent.

5[✓]: 3 marks

4[✓]: 2 marks

3[✓]: 1 mark

- (c) The following equation represents the rate of reaction between 2-bromo-2-methylpropane and aqueous sodium hydroxide under reflux.



Experiment	Initial $[(\text{CH}_3)_3\text{CBr}] / \text{mol dm}^{-3}$	Initial $[\text{OH}^-] / \text{mol dm}^{-3}$	Initial rate of reaction / $\text{mol dm}^{-3} \text{ s}^{-1}$
1	1.0×10^{-3}	1.0×10^{-1}	0.5×10^{-3}
2	2.0×10^{-3}	1.0×10^{-1}	1.0×10^{-3}
3	2.0×10^{-3}	0.5×10^{-1}	1.0×10^{-3}

[11]

- (i) Determine the order of reaction with respect to the two reactants. Hence, write the rate equation.

Comparing experiments 1 and 2

Initial rate of reaction is doubled when initial $[(\text{CH}_3)_3\text{CBr}]$ is doubled

Order of reaction with respect to $(\text{CH}_3)_3\text{CBr}$ is 1 [1]

Comparing experiments 1 and 3

Initial rate of reaction stays the same when initial $[\text{OH}^-]$ is halved

Order of reaction with respect to OH^- is 0 [1]

Rate = $k[(\text{CH}_3)_3\text{CBr}]$ [1]

- (ii) Calculate the value of the rate constant, stating its units clearly.

Using experiment 1

$$0.5 \times 10^{-3} = k \times 1.0 \times 10^{-3}$$

$$k = 0.500 \text{ s}^{-1} \quad [1]$$

- (iii) Calculate the initial rate of reaction when the initial concentration of $(\text{CH}_3)_3\text{CBr}$ is $2.50 \times 10^{-3} \text{ mol dm}^{-3}$ and the initial concentration of OH^- is $2.0 \times 10^{-1} \text{ mol dm}^{-3}$.

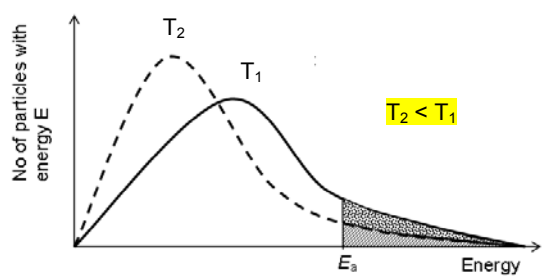
$$\text{Rate} = 0.500 \times 2.50 \times 10^{-3}$$

$$= 1.25 \times 10^{-3} \text{ mol dm}^{-3} \text{ s}^{-1} \quad [1]$$

- (iv) How would you expect the initial rate of reaction to change when 2-bromo-2-methylpropane is replaced by 2-chloro-2-methylpropane? Explain your answer.

The initial rate of reaction will be lower,[1] due to the breakage of the stronger C-Cl bond.[1]

- (v) With the aid of a Maxwell-Boltzmann distribution curve, explain what happens to the rate of reaction if the reaction mixture is surrounded by ice-cold water?



Correct shape of graph [✓]

Correct labeling of axes [✓]

At a lower temperature, [✓] there is a smaller number of reactant particles with energy greater than or equal to the activation energy, [✓] resulting in a decrease in the frequency of effective collisions [✓] between the reactant particles and hence a lower rate of reaction. [✓]

6[✓]: 4 marks

5[✓]: 3 marks

3-4[✓]: 2 marks

1-2[✓]: 1 mark

[Total: 20]

2. (a) A reaction scheme starting from methylbenzene is shown below. Draw the structures of the organic products A to E and state the reagents and conditions needed for steps I to V.

[6]

Formatted: Font: Bold

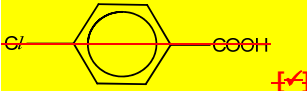
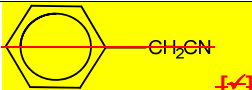
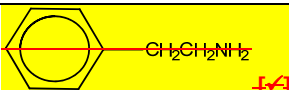
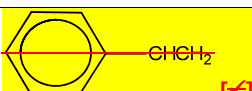
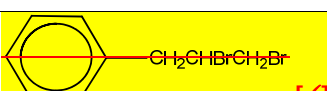
Step I: Cl_2 , Fe/ FeCl_3 / AlCl_3 catalyst [✓]

Step II: Cl_2 , uv light [✓]

Step III: ethanolic KCN, heat [✓]

Step IV: $\text{CH}_3\text{CH}_2\text{OH}$, a few drops of conc H_2SO_4 , heat under reflux [✓]

Step V: LiAlH₄ in dry ether [✓]

A:	 [✓]
B:	 [✓]
C:	 [✓]
D:	 [✓]
E:	 [✓]

2[✓]: 1 mark

(b) Rank the following compounds in order of increasing acidity:

Propanoic acid, propanol, 2-chloropropanol, 2-bromopropanoic acid

[4] Formatted: Centered

Acidity: ~~Propanol < 2-chloropropanol < propanoic acid < 2-bromopropanoic acid~~ [1]

Acid strength depends on stability of anion. The more stable the anion, the stronger the acid.

In 2-chloropropanol, the ~~electron-withdrawing Cl atom disperses the negative charge on the alkoxide anion~~, [✓] stabilising the anion. On the other hand, in propanol, the ~~electron-donating alkyl group intensifies the negative charge on the alkoxide anion~~, [✓] destabilising the anion.

In propanoic acid, the ~~negative charge on the carboxylate anion is delocalised into the COO⁻ group~~, [✓] stabilising the anion to a larger extent as compared to the alkoxide anion, [✓] In 2-bromopropanoic acid, the ~~negative charge on the carboxylate anion is both delocalised into the COO⁻ group and dispersed by the electron-withdrawing Br atom~~, [✓] thus stabilising the anion to the largest extent.

5[✓]: 3 marks

4[✓]- 2 marks

3[✓]- 1 mark

- (c) The following equation represents the rate of reaction between 2-bromo-2-methylpropane and aqueous sodium hydroxide under reflux.



Experiment	Initial $[(\text{CH}_3)_3\text{CBr}] / \text{mol dm}^{-3}$	Initial $[\text{OH}^-] / \text{mol dm}^{-3}$	Initial rate of reaction / $\text{mol dm}^{-3} \text{s}^{-1}$
1	1.0×10^{-3}	1.0×10^{-1}	0.5×10^{-3}
2	2.0×10^{-3}	1.0×10^{-1}	1.0×10^{-3}
3	2.0×10^{-3}	0.5×10^{-1}	1.0×10^{-3}

[4]

- (i) Determine the order of reaction with respect to the two reactants. Hence, write the rate equation.

Comparing experiments 1 and 2

Initial rate of reaction is doubled when initial $[(\text{CH}_3)_3\text{CBr}]$ is doubledOrder of reaction with respect to $(\text{CH}_3)_3\text{CBr}$ is 1 [1]

Comparing experiments 1 and 3

Initial rate of reaction stays the same when initial $[\text{OH}^-]$ is halvedOrder of reaction with respect to OH^- is 0 [1]Rate = $k[(\text{CH}_3)_3\text{CBr}]$ [1]

- (ii) Calculate the value of the rate constant, stating its units clearly.

Using experiment 1

$$0.5 \times 10^{-3} = k \times 1.0 \times 10^{-3}$$

$$k = 0.500 \text{ s}^{-1} \text{ [1]}$$

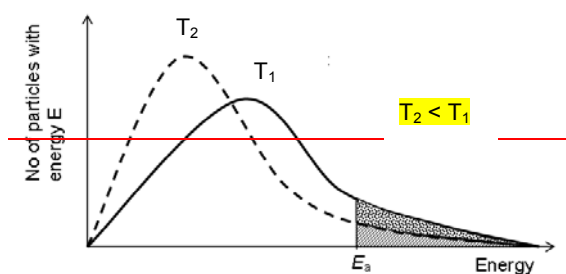
- (iii) Calculate the initial rate of reaction when the initial concentration of $(\text{CH}_3)_3\text{CBr}$ is $2.50 \times 10^{-3} \text{ mol dm}^{-3}$ and the initial concentration of OH^- is $2.0 \times 10^{-1} \text{ mol dm}^{-3}$.

$$\begin{aligned} \text{Rate} &= 0.500 \times 2.50 \times 10^{-3} \\ &= 1.25 \times 10^{-3} \text{ mol dm}^{-3} \text{ s}^{-1} \text{ [1]} \end{aligned}$$

- (iv) How would you expect the initial rate of reaction to change when 2-bromo-2-methylpropane is replaced by 2-chloro-2-methylpropane? Explain your answer.

The initial rate of reaction will be lower, [1] due to the breakage of the stronger C-Cl bond. [1]

- (v) With the aid of a Maxwell-Boltzmann distribution curve, explain what happens to the rate of reaction if the reaction mixture is surrounded by ice-cold water?



Correct shape of graph [✓]
 Correct labeling of axes [✓]

At a lower temperature, [✓] there is a smaller number of reactant particles with energy greater than or equal to the activation energy, [✓] resulting in a decrease in the frequency of effective collisions, [✓] between the reactant particles and hence a lower rate of reaction, [✓]

6[✓]: 4 marks
 5[✓]: 3 marks
 3-4[✓]: 2 marks
 1-2[✓]: 1 mark

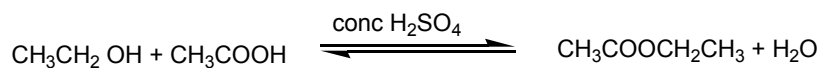
[Total: 20]

Formatted: Font: Bold

- 7 Esters are derivatives of carboxylic acids and are widely found in nature. Esters are also widely used in industry.

[10]

- (a) Ethyl ethanoate is present in wine and is also widely used as a solvent. Ethyl ethanoate can be synthesised from ethanol and ethanoic acid in the presence of concentrated sulfuric acid.



Formatted Table

- (i) Write an expression for K_c for the above reaction and state its units.

$$K_c = \frac{[\text{CH}_3\text{CO}_2\text{CH}_2\text{CH}_3][\text{H}_2\text{O}]}{[\text{CH}_3\text{CO}_2\text{H}][\text{CH}_3\text{CH}_2\text{OH}]} \text{ no units}$$

- (ii) A student carried out an experiment to determine K_c for the above reaction. 12.00 g of glacial ethanoic acid, 9.20 g of ethanol were reacted together in the presence of concentrated sulfuric acid. At equilibrium, the yield of ethyl ethanoate was 33.5% ~~and the total volume of the reaction mixture was 44.0 cm³.~~

Calculate the value for K_c .

$$\text{Initial amount of ethanoic acid} = 12.00 / 60.0 = 0.200 \text{ mol}$$

$$\text{Initial amount of ethanol} = 9.2 / 46.0 = 0.200 \text{ mol}$$

$$\text{Equilibrium amount of ethyl ethanoate} = 33.5/100 \times 0.200 = 0.0670 \text{ mol}$$

$$\text{Hence equilibrium amount of water} = 0.0670 \text{ mol}$$

$$\text{Equilibrium amount of ethanoic acid} = \text{Equilibrium amount of ethanol}$$

$$= 0.200 - 0.0670 = 0.133$$

$$K_c = (0.0670)^2 / (0.133)^2 = 0.254$$

- (iii) K_c for the above reaction increases with increasing temperature. State and explain if forward reaction is endothermic or exothermic.

K_c increasing with increasing temperature implies that the forward reaction is favoured. An increase in temperature favours an endothermic reaction to absorb the increase in heat. Thus the forward reaction is endothermic.

- (iv) A student repeated the above experiment but used an ethanol-water mixture instead of pure ethanol. Explain how the yield of the reaction will be affected.

Using an ethanol-water mixture increases the concentration of water and decrease the concentration of ethanol. The position of equilibrium is shifted to the left. Yield of the reaction is decreased.

- (v) The concentrated sulfuric used in the reaction has two roles. State these two roles.

1) Catalyst

2) Dehydrating agent

- (b) When an ester, A, $\text{C}_{10}\text{H}_{18}\text{O}_4$, is heated under reflux with dilute aqueous acid, B, $\text{C}_2\text{H}_4\text{O}_4$, and C, $\text{C}_4\text{H}_{10}\text{O}$, are produced.

Both B and C react with sodium metal to produce a colourless gas.

When B is heated under reflux with acidified potassium manganate(VII), a colourless gas is evolved. No organic product is produced.

C reacts with acidified potassium manganate(VII) to produce D.

Both C and D react separately with aqueous alkaline iodine to produce a yellow

Formatted Table

Formatted: Highlight

Formatted: Font: (Default) Arial, 11 pt, Italic, Font color: Red, Complex Script Font: Arial, Italic, Lowered by 15 pt, Highlight

Formatted: Font: Not Italic, Highlight

Field Code Changed

Formatted Table

Formatted: Not Superscript/ Subscript

Formatted: Highlight

Formatted: Highlight

Formatted Table

Formatted: Highlight

Formatted: Font: Not Italic, Highlight

Formatted Table

Formatted: Highlight

Formatted Table

Formatted: Highlight

Formatted: Bullets and Numbering

Formatted Table

precipitate.

D reacts with 2,4-dinitrophenylhydrazine to produce, E.

Deduce the structures of A, B, C, D and E and explain the chemistry of the reactions involved.

A undergoes acidic hydrolysis to produce B, a diacid and C, an alcohol.

B and C undergoes acid metal reaction with sodium metal. OH group present.

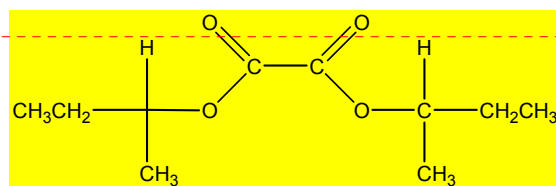
B undergoes oxidation with acidified potassium manganate to produce carbon dioxide. B is ethanedioic acid.

C undergoes oxidation with acidified potassium manganate. C is an alcohol.

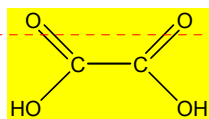
C undergoes oxidation with aqueous alkaline iodine. C has structure $\text{CH}_3\text{CH}(\text{OH})$ present.

D undergoes oxidation with aqueous alkaline iodine. D has structure $\text{CH}_3\text{C}=\text{O}$ present.

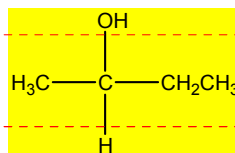
D undergoes condensation reaction with 2,4-DNPH. D is a ketone.



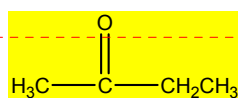
A



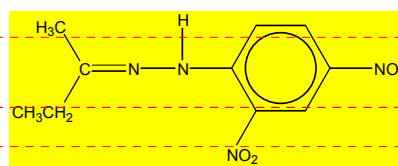
B



C



D



E

[Total: 20]

END OF PAPER

Formatted	... [1]
Formatted	... [2]
Formatted	... [3]
Formatted	... [4]
Formatted	... [5]
Formatted	... [6]
Formatted	... [7]
Formatted	... [8]
Formatted	... [9]
Formatted	... [10]
Formatted	... [11]
Formatted	... [12]
Formatted	... [13]
Formatted	... [14]
Formatted	... [15]
Formatted	... [16]
Formatted	... [17]
Formatted	... [18]
Formatted	... [19]
Formatted	... [20]
Formatted	... [21]
Formatted	... [22]
Formatted	... [23]
Formatted	... [24]
Formatted	... [25]
Formatted	... [26]
Formatted	... [27]
Formatted	... [28]
Formatted	... [29]
Formatted	... [30]
Formatted	... [31]
Formatted	... [32]
Formatted	... [33]
Formatted	... [34]
Formatted	... [35]
Formatted	... [36]
Formatted	... [37]
Formatted	... [38]
Formatted	... [39]
Formatted	... [40]
Formatted	... [42]
Formatted	... [43]
Formatted	... [44]
Formatted	... [45]
Formatted	... [46]
Formatted	... [47]
Formatted	... [49]
Formatted	... [50]
Formatted	... [48]
Formatted Table	... [51]
Formatted	... [52]

Page 24: [1] Formatted	S7000770C	9/27/2011 9:41:00 AM
Font: Bold, Highlight		
Page 24: [2] Formatted	S7000770C	9/27/2011 9:41:00 AM
Highlight		
Page 24: [3] Formatted	S7000770C	9/27/2011 9:41:00 AM
Font: Bold, Highlight		
Page 24: [4] Formatted	S7000770C	9/27/2011 9:41:00 AM
Highlight		
Page 24: [5] Formatted	S7000770C	9/27/2011 9:41:00 AM
Font: Bold, Highlight		
Page 24: [6] Formatted	S7000770C	9/27/2011 9:41:00 AM
Highlight		
Page 24: [7] Formatted	S7000770C	9/27/2011 9:41:00 AM
Font: Bold, Highlight		
Page 24: [8] Formatted	S7000770C	9/27/2011 9:41:00 AM
Highlight		
Page 24: [9] Formatted	S7000770C	9/27/2011 9:41:00 AM
Font: Bold, Highlight		
Page 24: [10] Formatted	S7000770C	9/27/2011 9:41:00 AM
Highlight		
Page 24: [11] Formatted	S7000770C	9/27/2011 9:41:00 AM
Font: Bold, Highlight		
Page 24: [12] Formatted	S7000770C	9/27/2011 9:41:00 AM
Highlight		
Page 24: [13] Formatted	S7000770C	9/27/2011 9:41:00 AM
Font: Bold, Highlight		
Page 24: [14] Formatted	S7000770C	9/27/2011 9:41:00 AM
Highlight		
Page 24: [15] Formatted	S7000770C	9/27/2011 9:41:00 AM
Font: Bold, Highlight		
Page 24: [16] Formatted	S7000770C	9/27/2011 9:41:00 AM
Highlight		
Page 24: [17] Formatted	S7000770C	9/27/2011 9:41:00 AM
Font: Bold, Highlight		
Page 24: [18] Formatted	S7000770C	9/27/2011 9:41:00 AM
Highlight		
Page 24: [19] Formatted	S7000770C	9/27/2011 9:41:00 AM
Font: Bold, Highlight		
Page 24: [20] Formatted	S7000770C	9/27/2011 9:41:00 AM

Highlight

Page 24: [21] Formatted	S7000770C	9/27/2011 9:41:00 AM
-------------------------	-----------	----------------------

Font: Bold, Highlight

Page 24: [22] Formatted	S7000770C	9/27/2011 9:41:00 AM
-------------------------	-----------	----------------------

Highlight

Page 24: [23] Formatted	S7000770C	9/27/2011 9:41:00 AM
-------------------------	-----------	----------------------

Font: Bold, Highlight

Page 24: [24] Formatted	S7000770C	9/27/2011 9:41:00 AM
-------------------------	-----------	----------------------

Highlight

Page 24: [25] Formatted	S7000770C	9/27/2011 9:42:00 AM
-------------------------	-----------	----------------------

Font: Bold, Highlight

Page 24: [26] Formatted	S7000770C	9/27/2011 9:41:00 AM
-------------------------	-----------	----------------------

Highlight

Page 24: [27] Formatted	S7000770C	9/27/2011 9:42:00 AM
-------------------------	-----------	----------------------

Font: Bold, Highlight

Page 24: [28] Formatted	S7000770C	9/27/2011 9:41:00 AM
-------------------------	-----------	----------------------

Highlight

Page 24: [29] Formatted	S7000770C	9/27/2011 9:42:00 AM
-------------------------	-----------	----------------------

Font: Bold, Highlight

Page 24: [30] Formatted	S7000770C	9/27/2011 9:41:00 AM
-------------------------	-----------	----------------------

Highlight

Page 24: [31] Formatted	S7000770C	9/27/2011 9:42:00 AM
-------------------------	-----------	----------------------

Highlight

Page 24: [32] Formatted	S7000770C	9/27/2011 9:42:00 AM
-------------------------	-----------	----------------------

Highlight

Page 24: [33] Formatted	S7000770C	9/27/2011 9:42:00 AM
-------------------------	-----------	----------------------

Highlight

Page 24: [34] Formatted	S7000770C	9/27/2011 9:42:00 AM
-------------------------	-----------	----------------------

Font: (Default) Arial, Bold, Complex Script Font: Arial, Highlight

Page 24: [35] Formatted	S7000770C	9/27/2011 9:24:00 AM
-------------------------	-----------	----------------------

Centered

Page 24: [36] Formatted	S7000770C	9/27/2011 9:42:00 AM
-------------------------	-----------	----------------------

Font: Bold, Highlight

Page 24: [37] Formatted	S7000770C	9/27/2011 9:42:00 AM
-------------------------	-----------	----------------------

Highlight

Page 24: [38] Formatted	S7000770C	9/27/2011 9:28:00 AM
-------------------------	-----------	----------------------

Centered

Page 24: [39] Formatted	S7000770C	9/27/2011 9:42:00 AM
-------------------------	-----------	----------------------

Highlight

Page 24: [40] Formatted	S7000770C	9/27/2011 9:42:00 AM
Highlight		
Page 24: [41] Formatted	S7000770C	9/27/2011 9:42:00 AM
Font: Bold, Highlight		
Page 24: [42] Formatted	S7000770C	9/27/2011 9:42:00 AM
Highlight		
Page 24: [43] Formatted	S7000770C	9/27/2011 9:42:00 AM
Highlight		
Page 24: [44] Formatted	S7000770C	9/27/2011 9:42:00 AM
Highlight		
Page 24: [45] Formatted	S7000770C	9/27/2011 9:28:00 AM
Centered		
Page 24: [46] Formatted	S7000770C	9/27/2011 9:42:00 AM
Highlight		
Page 24: [47] Formatted	S7000770C	9/27/2011 9:42:00 AM
Highlight		
Page 24: [48] Formatted	S7000770C	9/27/2011 9:42:00 AM
Font: Bold, Highlight		
Page 24: [49] Formatted	S7000770C	9/27/2011 9:42:00 AM
Highlight		
Page 24: [50] Formatted	S7000770C	9/27/2011 9:42:00 AM
Highlight		
Page 24: [51] Formatted Table	S7000770C	9/7/2011 2:54:00 PM
Formatted Table		
Page 24: [52] Formatted	S7000770C	9/7/2011 4:05:00 PM
Font: Bold		